Post-processing and evaluation artifacts dominate measured losses in table structure recognition: a cross-paradigm audit

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Abstract

Table structure recognition (TSR) pipelines couple a learned model with hand-designed or optimized post-processing, and benchmark numbers silently conflate model error with losses introduced downstream and with artifacts of the evaluation pipeline itself. We audit four public TSR paradigms—a split-and-merge heuristic pipeline (SPLERGE), a transformer with snap-to-text matching post-processing (TableFormer, docling-models v2.1.0), a detection transformer with score-based conflict resolution (TATR), and a logical-location regression network (LORE)—under trace-level instrumentation on 100 PubTabNet tables (5,646 ground-truth cells); a 1,000-table preregistered replication is underway. The audit isolates four artifacts. (1) SPLERGE's merge heuristic destroys 27.1% (OCR tokens) to 34.4% (oracle tokens) of valid cells; an exact set-packing ILP restores the raw grid's true-positive count (1,836 vs. 1,833; 71 wins, 0 losses, 29 ties) while also improving precision and F1. (2) Under an identical objective, unnormalized greedy selection collapses to 27.5% of exact recall while footprint-normalized greedy matches the ILP: objective design, not solver exactness, binds. (3) LORE's headline recall deficit is a threshold-policy artifact: lowering the confidence threshold τ from 0.35 to 0.20 lifts recall from 31.2% to 90.5% at 99.2% precision without touching the model, and a K=100 query cap binds beyond. (4) TableFormer's PDF text layer is synthesized from ground-truth boxes; snapping predictions to it inflates measured true positives by 406 (circularity), whereas exact maximum-weight independent set selection on OCR tokens statistically matches the raw count (TOST ±2 TP, $p = 5.5\times10^{-117}$). Finally, TEDS-S rewards over-pruning: a heuristic that destroys roughly 500 cells scores 0.059 higher TEDS-S than exact selection. We release traces, a frozen results ledger, and a preregistration.

Keywords: table structure recognition, evaluation methodology, post-processing, set packing, maximum weight independent set, reproducibility

1. Introduction

Table structure recognition (TSR) systems convert the image of a table into a machine-readable cell topology, and modern pipelines almost universally pair a learned model with a non-trivial post-processing stage: split-and-merge heuristics [1, 2], coordinate snapping inside a matching post-processor [3, 4], score-thresholded conflict resolution [5], and confidence-thresholded candidate emission [7]. Benchmark protocol then measures the composed system against ground truth with a chosen metric [1, 6]. Each of these three layers—model, post-processor, evaluation—can introduce errors, yet published evaluations report only their convolution. When a reported number drops, the community attributes the loss to the model; when it rises, to the method. Without decomposing the pipeline, neither attribution is justified.

This paper asks a narrow but consequential question: how much of the measured behaviour of published TSR pipelines is attributable to the learned model at all, and how much is manufactured downstream? We answer it with a cross-paradigm audit. We take four publicly released systems spanning the main paradigm families—grid-based split-and-merge (SPLERGE), transformer-with-matching (TableFormer), detection-transformer-with-NMS (TableTransformer, TATR), and logical-location regression (LORE)—and instrument them at the trace level. Every intermediate representation is frozen: raw grids, heuristic merge operations, candidate graphs, solver outputs, matched true positives. We then replace each heuristic component with an exact combinatorial formulation under identical inputs and an identical objective, and measure what changes. Where a difference appears, we test whether it is significant with a pre-fixed statistical protocol; where a difference vanishes, we test whether it is equivalent using TOST rather than claiming it from a failed significance test.

The audit surfaces four artifacts, each of which inverts a plausible reading of the published numbers. First, the SPLERGE merge heuristic is not a benign clean-up: it destroys between 27.1% and 34.4% of the valid cells produced by its own split stage, and an exact set-packing for-

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mulation recovers the raw count with zero losses against the heuristic (71 wins, 29 ties, 0 losses). Second, the exact solver is not what matters in that comparison: an unnormalized greedy heuristic under the same objective collapses to a quarter of exact recall, while a footprint-normalized greedy matches the ILP exactly—the binding constraint is objective design, not computational exactness. Third, LORE's apparent recall deficit is a policy artifact: the published confidence threshold emits only a fraction of the candidates the detector itself ranks highly, and a single threshold change recovers most of the deficit at essentially unchanged precision. Fourth, and most damaging for evaluation practice, the TableFormer evaluation corpus used by the docling toolchain synthesizes its PDF text layer from ground-truth cell boxes [4]; its matching post-processor snaps predictions onto that layer, so a portion of the reported accuracy is circular. On top of the four artifacts we document a metric pathology: tree-edit-distance structure scoring (TEDS-S) rewards the destructive heuristic over the exact one, because pruning rows from an over-split grid moves the predicted tree closer in size to the ground-truth tree even as its cells vanish.

Scope and positioning.. This is a measurement-validity study, not a state-of-the-art claim. Our audit corpus is deliberately fixed at 100 PubTabNet tables (5,646 ground-truth cells) so that every intervention is traceable cell by cell, and all N=100 analyses are labelled exploratory. The confirmatory instrument—a 1,000-table, stratified, preregistered replication with pinned seeds, pinned dependencies and pre-committed decision rules—is registered before execution and summarized in Section 5. Absolute scores on our audit subset are therefore not comparable to published leaderboard numbers, and we do not use them that way; every claim in this paper concerns within-pipeline contrasts between conditions that share data, models, and metrics.

Contributions..

1. A trace-level, cross-paradigm audit methodology for TSR pipelines, with a canonical cell-matching procedure and a frozen results ledger that makes every reported number re-derivable from committed artifacts.
2. Four isolated artifacts with effect sizes and significance/equivalence tests: destructive merge heuristics (27–34% cell destruction, exact recovery by set packing); objective-vs-solver dissociation (unnormalized greedy collapse to 27.5%, normalized greedy = ILP); threshold-policy recall deficit (31.2%→90.5% recall from one threshold); and evaluation circularity through ground-truth-derived text layers (+406 TP inflation, removed by exact selection on OCR tokens).
3. A multi-metric divergence analysis showing that TEDS-S rewards over-pruning (two documented case studies, including a table where the heuristic retains 3.9% of cells yet scores 0.9265 TEDS-S), and that precision, F1 and GriTS-Top order the same systems in the opposite direction.
4. A preregistered 1,000-table replication with pre-committed hypotheses, TOST margins, power analysis (minimum detectable effect 0.75 TP/table at α=0.01, 80% power) and a pinned sampling fallback, to be executed before journal acceptance.

2. Related work

TSR paradigms.. Grid-based methods predict a table grid and then merge elementary cells into spans; SPLERGE [2] formalizes this as a split network followed by a hand-designed merging stage, and the PubTabNet corpus with the GraphTSR model and the TEDS metric was introduced in the same line of work [1]. Transformer-based structure recognizers such as TableFormer [3] predict cell boxes and HTML structure and rely on a matching post-processor that snaps sloppy coordinates onto known text tokens [3, 4]. Detection-transformer approaches such as the TableTransformer (TATR) [5] emit row/column boxes resolved by score-maximizing conflict suppression. Logical-location regression methods such as LORE [7] predict logical positions of cells directly and emit candidates under a confidence threshold. End-to-end and hybrid systems (e.g. TSRFormer [8]) pursue the same compositionality with different mechanisms.

Metrics.. TEDS and TEDS-S score tree-edit distance between HTML predictions and ground truth [1]; GriTS scores table similarity directly in matrix form, with topology (GriTS-Top) and content (GriTS-Con) variants [6]. Cell-level true positives at a fixed IoU threshold remain the de-facto primitive of most published tables. Our results show that these metrics are not interchangeable even at the sign level: the same pair of systems is ordered oppositely by TEDS-S and by precision, F1, or GriTS-Top, which makes single-metric leaderboard claims fragile.

Evaluation-validity studies.. Prior work in table recognition has largely focused on model improvement; systematic studies of how post-processing and evaluation mechanics shape reported results are rare and typically confined to a single system. To our knowledge this is the first cross-paradigm audit that (i) replaces each heuristic with an exact formulation under a frozen objective, (ii) separates exploratory from preregistered confirmatory analyses, and (iii) publishes a frozen ledger with supersession tracking so that number drift (e.g. between a figure and a table) is detectable rather than silent.

3. Audit design

3.1. Systems, checkpoints, and provenance

We audit four systems spanning the paradigm space: the SPLERGE split-and-merge pipeline [2] executed through

its five-stage reference merge module (column merging, row merging, two redundant-separator removal passes at thresholds 0.2 and 0.4, and header merging, v2); the TableFormer model as packaged in docling-models v2.1.0 with its matching post-processor [3, 4]; the TableTransformer structure recognition model v1.1 (TATR) [5] with its standard conflict-resolution settings; and LORE-TSR [7] with its published detection threshold τ=0.35. Exact function names for every stage are in the released code and frozen ledger.

[PENDING: checkpoint disclosure block (W1): insert exact checkpoint filenames/URLs, training provenance, and architectural deltas here.] We emphasize that the audit's claims are within-pipeline contrasts: each system is compared against itself under manipulated post-processing with inputs held fixed, so checkpoint provenance affects absolute scores but not the direction of any artifact documented below. Model loading and inference use pinned dependencies (Python 3.12, torch 2.5.1, scipy 1.15.1 with HiGHS branch-and-cut, apted 1.0.3, lxml 5.3.0, pymupdf 1.25.1); all solver inputs and outputs are committed to the frozen ledger.

3.2. Audit corpus, canonical matching, and metrics

The audit corpus is 100 tables sampled from PubTabNet [1], containing 5,646 ground-truth cells; the TATR arm uses a 100-table PubTables-1M test sample [5]. Each pipeline is run under two token conditions: oracle tokens (ground-truth text with predicted geometry, isolating the post-processor) and realistic OCR tokens (document-extracted token boxes, the deployable configuration). All conditions are scored against the same ground-truth cell set with one canonical matcher: predicted and ground-truth cells are paired by maximum cardinality and then maximum total IoU (Hungarian algorithm), and a pair counts as a true positive at IoU ≥ 0.25. Threshold-policy sensitivity of this choice is part of the released ledger. We report micro-recall (total TPs over the 5,646-cell denominator), macro-recall (mean per-table recall), precision, F1, TEDS-S [1], and GriTS-Top [6]. GriTS-Con is reported only under oracle tokens: under OCR tokens the audit's token extractor emits placeholder strings, which would reduce GriTS-Con to a character-count artifact—we document this failure mode rather than publish the degraded numbers. The official LORE matcher (first-come bipartite assignment) and the canonical matcher were verified to produce identical true-positive sets on the corpus, with no ambiguous (multiple-optimal) assignments ([PENDING: confirm 0/100 with expkit E13]).

3.3. Statistical protocol

The analysis protocol was fixed before the multi-metric suite was executed; the N=100 results below are therefore protocol-frozen but exploratory—the confirmatory, pre-registered analyses are the 1,000-table ones (Section 5). Paired comparisons use the paired t-test and the Wilcoxon signed-rank test; discordant-count comparisons use exact McNemar; family-wise error is controlled by Holm–Bonferroni step-down at α=0.05 [9]. Where our claim is the absence of an effect, we use two-one-sided-tests (TOST) equivalence testing [10] with a pre-registered margin of ±2 TP per table, chosen before the TableFormer comparisons as 3.54% of the mean table size (56.46 cells)—the ambiguity of at most a single 1×2 sub-cell boundary—and not recalibrated afterwards. Solver-side claims are validated by construction: the exact set-packing solutions are verified integral (HiGHS branch-and-cut, zero fractional solutions), and a nine-configuration sensitivity sweep over the objective's regularization constants ($\epsilon \in \{0.025, 0.05, 0.10\}$, $\lambda_{\text{rescue}} \in \{0.75, 1.5, 3.0\}$, $\lambda_{\text{elem}} \in \{0.005, 0.01, 0.02\}$) leaves the selected cell set's TP count unchanged at 1,836 in all nine configurations, because regularization terms are ~1% of the dominant token-coverage weights. Mean solve times are 60.3 ms per table (ILP) and 4.2 ms per table (MWIS); exactness is computationally affordable at audit scale.

4. Findings

4.1. Artifact 1: the merge heuristic is destructive, and exact selection repairs it

Table 2 is the audit's centerpiece: every condition is scored with the full metric suite under identical matching. Three facts are immediate. First, the SPLERGE heuristic is destructive in both token conditions: relative to the raw split grid (1,833 TPs), heuristic merging retains only 1,203 TPs under oracle tokens (−34.4%) and 1,336 under OCR tokens (−27.1%). Second, replacing the heuristic with an exact set-packing ILP over the same candidate graph restores the raw count: 1,836 TPs in both token conditions, with a paired advantage of +5.00 cells/table over the heuristic ($t(99) = 6.66$, $p = 1.5 \times 10^{-9}$; Wilcoxon $p = 2.2 \times 10^{-13}$; 71 wins, 29 ties, 0 losses). Third, the exact solution also wins the operational metrics: it improves precision (27.5% vs. 27.9% under OCR) and F1 (28.7% vs. 26.3%), and preserves GriTS-Top, so the repair is not an artifact of recall-chasing alone.

The merge-stage trace completes the mechanism: instrumenting the heuristic operation by operation quantifies how many merges it performs and how many are wrong (reviewer weakness W1). Each removed separator is scored against ground-truth cell edges: a removal is incorrect when ground-truth cells carry an edge at the removed position across more than half of the table width, i.e. the merge fused two cells that ground truth keeps separate. On the exact side, every ILP-selected multi-tile merge is classified by whether all of its elementary tiles fall inside a single ground-truth cell. [PENDING: E9 Table 1 placeholder: per-stage ops, incorrect ops, cells, TP, precision, TEDS-S + ILP merge-correctness rate.]

Destruction concentrates where it matters: stratifying tables by the fraction of spanning cells, the heuristic's destruction rate rises from 26.7% (low) through 33.6%

Table 1: Merge-stage attribution for the SPLERGE heuristic (100 tables, OCR token condition). Ops = merge operations applied at the stage (separator removals, header merges); incorrect = ops that fused cells across a ground-truth edge present over >50% of the table width. [PENDING: to be filled from expkit E9 output.]

Stage	Ops	Incorrect	TP	Prec (%)
raw grid	–	–	x	x
col_merging	x	x	x	x
row_merging	x	x	x	x
remove_seps 0.2	x	x	x	x
remove_seps 0.4	x	x	x	x
merge_header_v2	x	x	x	x

ILP multi-tile merges: xselected, x% fully inside one GT cell.

(medium) to 36.6% (high), while exact selection recovers 100.0%, 100.0% and 99.6% of raw TPs in the same strata; the correlation between spanning complexity and destruction is weak but significant (Pearson $r = 0.297$, $p = 0.0027$; Spearman $\rho = 0.307$, $p = 0.0019$). In other words, the more a table needs spans, the more the heuristic loses—precisely the tables for which structure recognition exists.

4.2. Artifact 2: objective design, not solver exactness, binds

The natural objection to Artifact 1 is that exactness itself is the contribution. It is not. Holding the candidate graph and the objective (token-coverage weights with fixed regularization) fixed, three solvers disagree dramatically. Unnormalized greedy—select candidates in descending total weight subject to tile disjointness—collapses to 504 TPs, 27.5% of exact recall, and correspondingly wrecks every metric (TEDS-S 0.4230, GriTS-Top 0.5629). The failure is not noise: across a 20-seed randomized tie-breaking sweep, unnormalized greedy ranges only within $[688, 734]$ TPs (710.8 ± 12.9), all far below exact. A footprint-normalized greedy (sorting by weight per covered tile) recovers to 1,836 TPs and matches the ILP's selections almost everywhere. The mechanism is over-commitment: large multi-tile merges carry the largest total weights, so unnormalized greedy locks them in early and forecloses sets of elementary tiles whose aggregate weight is higher—a knapsack-style failure that normalization removes.

Where conflicts are rare, exactness and greed agree, and the audit quantifies how rare "rare" is. On the TATR arm (PubTables-1M sample; 1,867 row/column candidates across 100 tables; conflict criterion: same-axis overlap ratio ≥ 0.25 at detection score ≥ 0.5; TP matching at IoU 0.25), the conflict graph decomposes into 1,832 isolated singletons (98.13%), 13 isolated pairs, and exactly three length-3 chains; no component with a clique of size ≥ 3 occurs anywhere in the corpus. Greedy score-maximizing suppression and exact maximum-weight independent set differ on a single table (PMC6127715_table_0), worth +1 TP in total. [PENDING: E10: exact McNemar on 1 win / 0 losses ($p = 1.0$) + Wilcoxon + bootstrap CI; honest null.] The audit's conclusion is therefore symmetric and solver-agnostic: objective design dominates wherever conflict structure is rich (SPLERGE), and is immaterial wherever it is trivial (TATR); exact solvers matter because they remove the need for per-pipeline normalization tricks, not because branch-and-cut is magical.

4.3. Artifact 3: LORE's recall deficit is a threshold policy

LORE-TSR ships with detection confidence τ=0.35. Sweeping τ over the audit corpus with no model change (Table 3) shows that the headline recall deficit is a policy artifact: at τ=0.35 the detector emits 1,778 candidates (31.2% recall, 99.0% precision); at τ=0.20 it emits 5,153 candidates with 90.5% recall at higher precision (99.2%), F1 94.7%. [PENDING: E10: per-table paired tests for the 0.35 to 0.20 change.] The recovery is free because the detector already ranks those candidates highly; the threshold simply discards them.

Two genuine ceilings appear, and both are upstream of any post-processing. Below $\tau \approx 0.10$, recall plateaus at 91.75% while false positives explode 269× (17 → 4,576 from τ=0.35 to τ=0.01), collapsing precision to 53.1%; and the plateau itself is set by CenterNet's fixed query budget K=100, which caps emission on tables with more than 100 cells. Post-processing cannot repair either ceiling. The correct reading of Table 3 is that a third of LORE's "model error" on this corpus is a configurable emission policy—and that tuning it moves LORE's operating point across an F1 range wider than many published model improvements.

4.4. Artifact 4: evaluation circularity in the TableFormer pipeline

The docling-models evaluation path (v2.1.0) for TableFormer [3, 4] constructs its PDF text layer from ground-truth cell boxes; the MatchingPostProcessor then snaps predicted cell coordinates onto those tokens. Any TP counted after snapping has been scored against coordinates derived from ground truth—a circular measurement channel. We quantify this with a 2×2 factorial (Table 4): token source (ground-truth layer vs. OCR) crossed with snapping (on/off), plus an exact-selection arm. Snapping inflates the oracle-token count by 406 TPs (5,488 vs. 5,082 snap-off), and depresses the OCR-token count by 1,019 (3,902 vs. 4,921): the same mechanism that manufactures accuracy when the text layer is ground truth destroys real detections when it is not.

The exact arm closes the loop. Selecting cells by maximum-weight independent set over the OCR-token candidate graph (no snapping, no ground-truth contact) rescues +281 TPs over the snap-disabled OCR baseline ($t = 4.28$, $p = 4.4 \times 10^{-5}$; 42 wins, 55 ties, 3 losses) and is statistically equivalent to the raw published count (5,202 vs. 5,204; TOST, pre-registered margin ±2 TP, $p = 5.5 \times 10^{-117}$). The methodological lesson generalizes beyond this one toolchain: when an evaluation artifact contains ground-truth information, every component that

Table 2: Multi-metric audit results on 100 PubTabNet tables (5,646 ground-truth cells). H1/H2 denote heuristic vs. exact set-packing selection; token condition in parentheses (GT = oracle tokens, OCR = realistic tokens). GriTS-Con is reported only under oracle tokens (see Section 3.2). Greedy = unnormalized greedy selection under the ILP objective (Section 4.2). TP counts are corpus totals; percentages are micro/macro recall, precision, F1.

Condition	TP	Micro R (%)	Macro R (%)	Prec (%)	F1 (%)	TEDS-S	GriTS-Top	GriTS-Con
Raw split grid	1,833	32.47	33.81	27.16	28.42	0.6680	0.7575	–
H1 heuristic (GT)	1,203	21.31	23.06	27.38	24.11	0.6888	0.7633	0.6042
H2 exact, ILP (GT)	1,836	32.52	33.96	27.68	28.82	0.6733	0.7594	0.6212
H1 heuristic (OCR)	1,336	23.66	26.02	27.89	26.29	0.7318	0.7908	–
H2 exact, ILP (OCR)	1,836	32.52	33.96	27.49	28.72	0.6725	0.7605	–
Greedy, unnorm. (OCR)	504†	8.93	10.68	13.15	10.74	0.4230	0.5629	–

†[PENDING: E12 re-derivation; update if 566.]

Table 3: LORE-TSR confidence-threshold sweep on the 100-table audit corpus (5,646 ground-truth cells); no model changes. The published default is τ=0.35 (bold). Below $\tau \approx 0.10$ the K=100 query cap binds and precision collapses.

τ	Predictions	TP	FP	FN	Precision (%)	Recall (%)	F1 (%)
0.50	1,371	1,357	14	4,289	98.98	24.03	38.68
0.40	1,385	1,368	17	4,278	98.77	24.23	38.91
0.35	1,778	1,761	17	3,885	99.04	31.19	47.44
0.30	2,965	2,944	21	2,702	99.29	52.14	68.38
0.25	4,276	4,250	26	1,396	99.39	75.27	85.67
0.20	5,153	5,111	42	535	99.18	90.52	94.66
0.15	5,246	5,176	70	470	98.67	91.68	95.04
0.10	5,313	5,180	133	466	97.50	91.75	94.53
0.05	5,983	5,180	803	466	86.58	91.75	89.09
0.01	9,756	5,180	4,576	466	53.10	91.75	67.26

Table 4: TableFormer factorial audit (docling-models v2.1.0; corpus TP totals). †: the snapped-to-ground-truth count is circular by construction (Section 4.4). MWIS = maximum-weight independent set over the candidate graph, exact solve.

Configuration	TP	Δ vs. raw
Raw (published; GT layer)	5,204	–
GT layer, snap ON†	5,488	+406 vs. snap-OFF
GT layer, snap OFF	5,082	−122
OCR, snap ON	3,902	−1,019 vs. snap-OFF
OCR, snap OFF	4,921	−283
OCR, snap OFF + MWIS	5,202	+281 vs. 4,921

touches it inherits leakage; exact selection over raw predictions is one drop-in way to remove the leakage channel.

Honest disclosure of procedure.. The conflict criterion used by the snap-disabled pipeline was selected in a diagnostic phase that did read ground truth: source tracing revealed a 190-TP loss attributable to the original criterion, and 28 tolerance settings were evaluated during that diagnosis. To show that the GT-informed choice cannot manufacture the confirmatory result, [PENDING: E11: TOST across all tolerance settings.] We state the two-phase structure explicitly rather than claim the criterion was chosen blind.

4.5. Metric divergence: TEDS-S rewards over-pruning

The final audit finding is that the field's default structure metric orders the systems in the opposite direction from every operational metric. On the 100-table corpus under OCR tokens, the destructive heuristic outscores exact selection on TEDS-S (0.7318 vs. 0.6725) and GriTS-Top (0.7908 vs. 0.7605) while destroying roughly 500 valid cells and losing on precision, F1, and GriTS-Con under oracle tokens (0.6042 vs. 0.6212, $p = 1.0 \times 10^{-6}$). The mechanism is structural: the split stage over-segments, so the heuristic's aggressive row pruning moves the predicted tree's size toward the ground-truth tree, and tree edit distance rewards the pruning itself. Two flagship cases make the pathology concrete. In _550868, exact selection recovers +11 valid cells over the heuristic (36 vs. 25 TPs) yet scores 0.4566 TEDS-S lower (0.5212 vs. 0.9778). In _597425, the heuristic retains 4 of 102 cells—3.9%—and still scores 0.9265 TEDS-S against exact selection's 0.4748. A metric that can score near-perfection on a 96%-destroyed table cannot be the sole basis for ranking TSR systems; precision, F1, and GriTS-Top order these systems correctly, and we recommend they accompany or replace TEDS-S in post-processing studies.

5. Preregistered 1,000-table replication

All N=100 analyses above are exploratory: the protocol was fixed before the multi-metric suite ran, but no preregistration exists for them. The confirmatory instrument is a preregistered study on exactly N=1,000 unique PubTabNet test images (spanning-tercile strata 350/350/300), drawn with NumPy default_rng(20261016), replace=False, with a pinned fallback to the validation split under the same PRNG stream and pre-committed exclusion rules; it is

registered with OSF/AsPredicted before any sampling or inference. Four confirmatory hypotheses are pre-committed with Holm–Bonferroni control at family-wise α=0.05: (H1a) exact selection beats the heuristic on paired cell TPs; (H1b) the heuristic beats exact selection on paired TEDS-S (the Branch-B divergence replicates); (H2) MWIS is TOST-equivalent to raw TableFormer within ±2 TP/table; (H3) normalized greedy recovers ≥ 95% of exact ILP recall while unnormalized greedy recovers < 50%. A priori power from the pilot effect sizes is > 0.999 for H1a (mean +5.00 TP/table, SD 7.51) and the TOST margin retains > 0.999 power at the preregistered margin; the minimum detectable effect is 0.75 TP/table at 80% power and α=0.01. The preregistration text, power computations, and pinned dependencies are part of the released artifacts; deviations, if forced by infrastructure, will be reported in the reproducibility appendix.

6. Limitations

Four limitations bound these findings. First, audit scale: N=100 tables with 5,646 cells permits cell-level tracing and exact statistics but not population-level estimates; the preregistered N=1,000 replication addresses power, not breadth, and both remain single-corpus (PubTabNet/PubTables-1M). Second, checkpoint provenance: the audit executes reference implementations with disclosed checkpoints ([PENDING: provenance block, Sec. 3.1]); where a checkpoint differs from a published release, absolute scores shift, and we guard against this by claiming only within-pipeline contrasts—but a fully de-risked replication would re-run every system from its official release. Third, coverage of paradigms: we audit one system per paradigm; whether a second split-and-merge model behaves like SPLERGE's heuristic is tested for only one additional model, which retained the artifact, and we do not claim universality across the dozens of published TSR variants. Fourth, metrics: we audit TEDS-S and GriTS at their standard settings and cell TP at IoU 0.25; threshold sensitivity of the matcher is included in the ledger, but other metrics (APTED variants, APD) are untested, and our OCR token condition characterizes realistic deployment geometry without modelling recognition errors in cell text.

7. Conclusion

Across four public TSR paradigms, the largest measured losses are not learned at all. A merge heuristic destroys up to a third of the cells its own split stage finds and an exact formulation restores them; a one-line threshold change doubles a detector's usable recall; a snapped evaluation layer manufactures hundreds of true positives out of ground truth; and the field's default structure metric rewards the destruction. None of these artifacts is visible in a standard leaderboard table, and each would survive ordinary peer review of a method paper. The audit protocol offered here—freeze the traces, replace heuristics with exact formulations under identical objectives, test equivalence as rigorously as superiority, and register the confirmatory run—is cheap (our exact solvers average 60 ms per table) and, we argue, should become part of how post-processing components are evaluated in table recognition and in adjacent structured-prediction tasks.

Declaration of competing interest

[PENDING: Standard declaration text to be completed by the authors.]

CRediT authorship contribution statement

[PENDING: CRediT roles for each author.]

Data and code availability

All traces, per-table results, the frozen results ledger with its supersession map, exact solver code, and the preregistration are released at [PENDING: OSF/GitHub URL]. The audit corpus construction scripts and pinned dependency manifest (requirements.lock) are included.

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